METRIC
A-A-52424B
August 3, 2011
SUPERSEDING
A-A-52424A
October 21, 1994

COMMERCIAL ITEM DESCRIPTION

CAPS, RADIATOR: PRESSURE RELIEVING (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers radiator pressure caps with a nominal pressure rating of 48 kPa. Caps covered by this CID are designed for use with industrial and automotive internal-combustion engine radiators.

2. CLASSIFICATION. Radiator caps covered by this CID are listed as follows:

Type	Design	Cap size
1	without handle	(64 mm)
2	with handle	(95 mm)

3. SALIENT CHARACTERISTICS

3.1 Materials. The materials used in the radiator cap shall be made of corrosion resistant steel with exception to the handle, crown, hook and chain which can be made of zinc plated steel per ASTM B633. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 4).

3.2 Chain and hook. Radiator cap shall be supplied with chain (see ASTM A466/ A466M class SF) and hooks. Hook design is optional but shall be easy to hook and remove without damage to the radiator cap, chain or hook (see figures 1 and 2).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 2930

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3.3 Compatibility. Radiator caps shall connect to standard filler necks conforming to the following:

Radiator Cap Type	Standard filler neck
1	A-A-52521 Type 1
2	A-A-52521 Type 2

3.4 Design and performance. Caps shall meet the requirements of this CID when used with antifreeze/coolant. Design shall be as specified in figures 1 and 2.

3.4.1 Fluid resistance. Gasket shall show no evidence of deterioration, flaking, change in volume (not more than 3%), or coolant discoloration when immersed in coolant such as antifreeze or alcohol for 158 hours.

3.4.2 Corrosion resistance. Caps shall conform to 2.4.3, 2.4.4 and 2.4.5 after being subjected to a salt spray test for 96 hours in accordance with ASTM B117. The salt spray shall not use ozone depleting substances, such as 1,1,1 Trichloroethane. Alternative methods or solvents shall be used in place of ozone depleting substances.

3.4.3 Pressure valve. Pressure valve shall relieve at no less than 45 kPa and no more than 55 kPa at a temperature of 68 degrees Celsius (°C).

3.4.4 By-passing. Pressure valves shall not show evidence of by-passing at pressures up to 43 kPa.

3.4.5 Vacuum valve. The vacuum valve shall relieve at less than 7 kPa vacuum at a temperature of 68 °C.

3.5 Extreme low temperature. The cap shall be able to be removed by hand from the stop contact points of the filler neck at the environmental temperature of -54 °C using arctic mittens. The torque used to remove the cap shall not be more than 5.6 newton meters (N·m) for type 1 and 14.1 N·m for type 2.

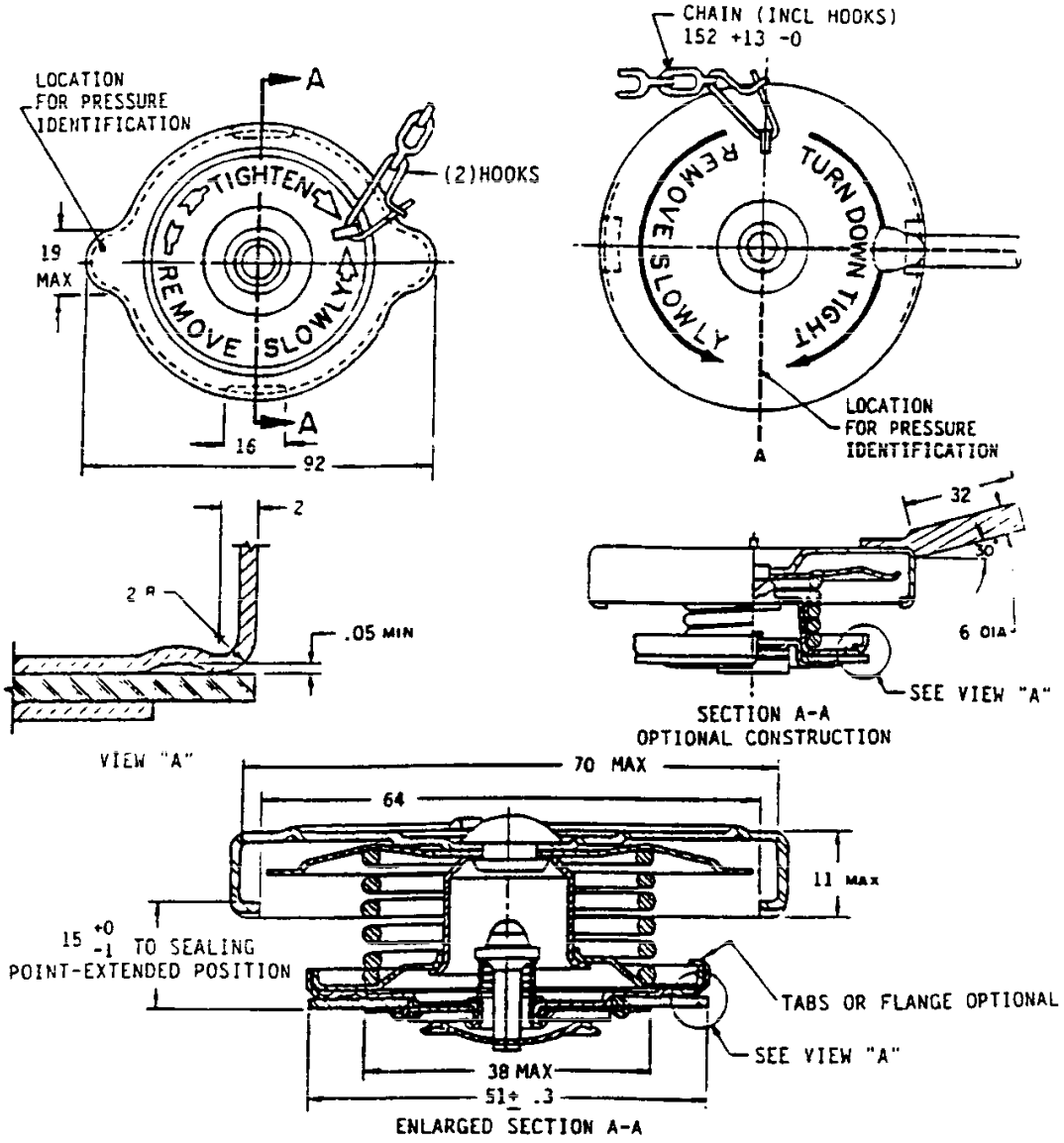
3.6 Identification and marking. Identification and marking of the radiator cap, as a minimum, shall include the part identification number, pressure identification, manufacturer's CAGE code (or equivalent identification), and part number. The radiator cap shall have markings to indicate direction of turn to remove or tighten the cap.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

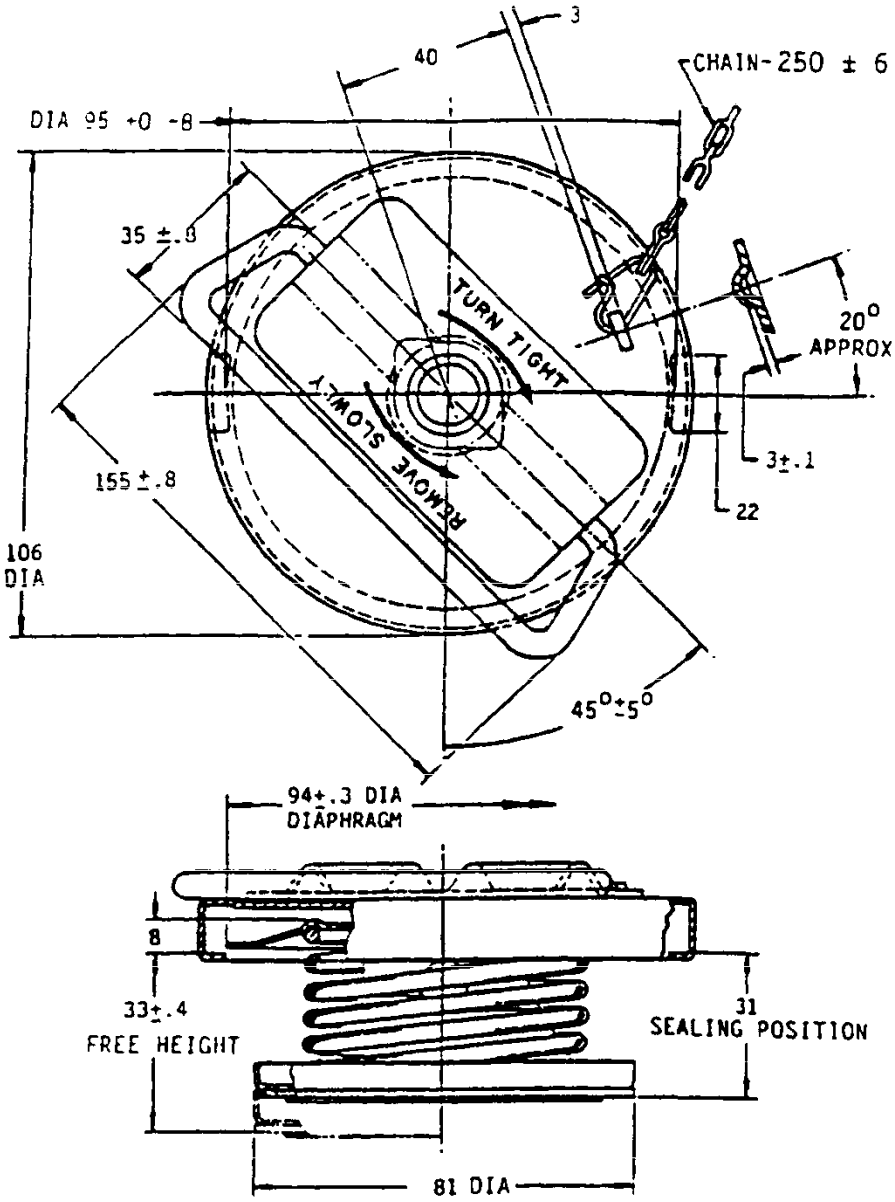


PIN No.	Former MS Part No.
A 52424-1	MS35840-1

NOTE: Dimensions are in millimeters. Tolerance ±.25

FIGURE 1. Radiator cap - type 1.

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PIN No.	Former MS Part No.
A 52424-2	MS51062-1

NOTE: Dimensions are in millimeters. Tolerance ±.25

FIGURE 2. Radiator cap - type 2.

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5.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.2).

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents.

7.1.1 Government specifications and standards. Copies of A-A-52521, Filler Neck: Radiator (Metric), are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

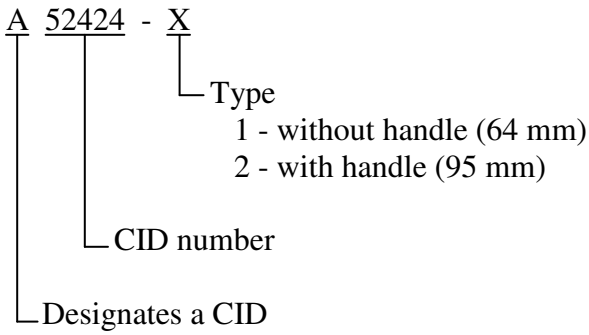
7.1.2 Non-Government publications. Copies of ASTM A466/ A466M "Standard Specification Weldless Carbon Steel Chain"; ASTM B117 "Standard Test Method of Salt Spray (Fog) Testing"; and ASTM B633 "Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel"; are available online at www.astm.org or from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. If required, the specific issue of individual documents referenced.
c. Selection of applicable level and packaging requirements (see 6).
d. Part or identification number.

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7.3 Part or identifying number (PIN). The PIN to be used for radiator caps acquired to this CID is as follows:



7.4 Cross-reference and supersession data. Radiator caps conforming to this CID are interchangeable/substitutable with radiator caps conforming to MIL-C-45167C, dated 20 April 1988, MS35840D, dated 21 January 1976, and MS51062E, dated 26 March 1979, which are superseded by this CID.

Custodians:
Army - AT
Navy - MC
Air Force - 99

Review Activity:
Army – CR4
DLA - CC

Preparing Activity:
Army - AT

(Project 2930-2009-003)

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